DO NOW — RECALL

Lesson 076 · Algorithm Choice: Time, Space and Precond

Name: _____ Class: _____ Date: _____

1. What must binary search assume?

2. Define auxiliary space.

3. Why can preprocessing change a choice?

4. Binary search in [1,3,5,7,9,11,13,15,17] for 15: give midpoint indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Data are sorted.

2. Model answer

Extra working memory beyond input storage.

3. Model answer

A one-time sort has cost.

4. Model answer

mid4=9, mid6=13, mid7=15; return index7.

By the end, you can…

• Choose an algorithm based on input properties, particularly whether data are already sorted.

• Compare running time AND auxiliary memory: linear scan O(n) time, O(1) extra; iterative binary search O(log n), O(1) extra on sorted data.

• Sorting first takes time; consider that one-time cost and the number of future searches.

Vocabulary — know these terms

precondition — Input requirement for correctness.

preprocessing — Preparing data before main operation.

amortise — Spread one-off cost across tasks.

NEW LEARNING

Read, annotate and explain

Understand

Choose an algorithm based on input properties, particularly whether data are already sorted.

Trace

Compare running time AND auxiliary memory: linear scan O(n) time, O(1) extra; iterative binary search O(log n), O(1) extra on sorted data.

Explain

Sorting first takes time; consider that one-time cost and the number of future searches.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Choose a method to find 7 in unsorted [4,7,2].

Check the working

Linear search: inspect4 then7, returning index1.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Choose linear/binary for unsorted [8,2,5,9], target5. [4]

Question 2

Trace binary search for 12 in [2,5,7,9,12,16,20]. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare one search in short unsorted list versus many searches in large fixed dataset. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Binary search must always be preferred even for one lookup in unsorted data.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.